

Solar-Powered Night-time Cooling Framework

Bachelor Thesis Project II
IIT Kharagpur



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Abstract

This project presents a dynamic, software-driven framework designed to resolve the temporal mismatch between peak renewable energy generation and high-intensity cooling demand at the Indian Institute of Technology Kharagpur. The campus infrastructure currently supports approximately 5.5 MWp of solar PV, generating an annual yield of 6 GWh. However, the residential cooling requirement in the Halls of Residence peaks during the midnight window, creating a significant reliance on the state power grid after active solar generation has ceased.

The proposed framework establishes an interactive methodology to "time-shift" daytime solar surplus into customizable night-time cooling for over 8,173 rooms across 21 Halls of Residence. Utilizing a granular analysis of three years of generation data (2023–2025), the system integrates a thermodynamic demand model with high-resolution parameter control. Key variables, including Cooling Intensity (0.1 to 1.0 TR/room), Chiller Efficiency (CoP 2 to 5), and Storage Scaling Factor ($X\%$), are adjustable in real-time through a dedicated management interface. The framework offers dual operational modes: a **Waterfall Tiered** logic that ensures campus-wide equity through a 2-3-4 hour waterfall discharge, and a **Manual Allocation** mode for granular control over individual hall durations. The project demonstrates that by optimizing these variables, the campus can achieve equitable thermal comfort using approximately 53.7% of the daily solar surplus, significantly reducing its carbon footprint and grid dependence.

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Chapter 1

Introduction

1.1 Project Rationale

The motivation for this framework stems from the unique climatic and infrastructural conditions of the IIT Kharagpur campus. Located in a region characterized by high ambient temperatures and significant humidity, the thermal comfort of students in the Halls of Residence is a primary institutional concern. While the institute has successfully deployed a massive 5.5 MWp solar PV network, the utilization of this energy is traditionally confined to daytime institutional loads, such as laboratories and administrative blocks.

Currently, centralized air conditioning infrastructure is limited to three high-density residences: SBP-1, SBP-2, and ABVP. For the remaining 19 halls, the implementation of such systems remains a proposal. However, the existing infrastructure already highlights significant energy management challenges that would only intensify under a full campus-wide deployment.

During the night, when the heat index remains high and the demand for residential cooling reaches its peak, the campus faces two major inefficiencies:

- **Underutilization of Renewable Surplus:** During weekends or vacation periods, a significant portion of daytime solar generation is underutilized or fed back into the grid at lower efficiency.
- **Projected Peak Grid Stress:** Even with current AC-equipped halls, the operation of centralized chillers between 10 PM and 4 AM creates a significant electrical surge. Scaling this centralized cooling model to all 21 Halls of Residence—as proposed in this project—would create a massive, unsustainable instantaneous load on the state power grid, leading to prohibitive operational costs and potential grid instability.

1.2 The Software-Driven Solution

This project establishes a comprehensive, data-driven framework to bridge this gap. By analyzing the "Safe Harvest" limits from the 2023–2025 solar datasets, the system calculates the exact surplus available for storage. Unlike static cooling models, this framework is built on variable thermodynamic parameters, allowing administrators to tune the system based on actual hardware performance, such as the Coefficient of Performance (CoP) of the chillers and the desired Tons of Refrigeration (TR) per room. This flexibility is essential for transitioning from the existing baseline halls to the broader campus proposal.

1.3 Framework Objectives

The primary objective of this project is to transition from a fixed energy model to a Dynamic Decision Support System (DSS) capable of managing both existing and proposed cooling loads. This is achieved through:

1. **Variable Parameter Modeling:** Allowing real-time adjustments to cooling density and system efficiency to reflect seasonal and mechanical variations.
2. **Multimodal Scheduling:** Implementing a **Waterfall Tiered** logic for campus-wide equity while providing a **Manual Allocation** for targeted energy dispatch.
3. **Sustainability Optimization:** Sizing storage systems at an optimal scaling factor to maximize solar utilization without cannibalizing daytime academic requirements.

By integrating these elements into a functional Python/Streamlit interface, this framework provides a transparent, scalable pathway for institutional sustainability and improved residential comfort at IIT Kharagpur.

Chapter 2

Solar Resource Assessment

2.1 Infrastructure Analysis: RESCO vs. CAPEX

The solar PV ecosystem at IIT Kharagpur is bifurcated into two distinct operational and financial models. Understanding the technical nuances between these two is vital for accurate generation forecasting.

2.1.1 CAPEX (Capital Expenditure) Model

The CAPEX installations (totaling 2500 kWp) are owned and operated directly by the institute.

- **Operational Control:** Since these assets are institutional property, there is greater flexibility in terms of maintenance schedules and integration with campus micro-grids.
- **Financial Profile:** The upfront cost was borne by the institute, but the per-unit cost of electricity generated is essentially the cost of maintenance.
- **Technical Observation:** CAPEX units on campus are often older installations that provide a stable baseline but may show slightly lower efficiency compared to the newer RESCO panels due to natural degradation.

2.1.2 RESCO (Renewable Energy Service Company) Model

The RESCO installations (totaling 2940 kWp) operate on a "Power Purchase Agreement" (PPA). A third-party company installs, owns, and maintains the panels, while the institute pays for the electricity consumed at a pre-decided tariff.

- **Modern Infrastructure:** RESCO installations at IITKGP are generally newer, utilizing high-efficiency mono-crystalline or polycrystalline modules with better performance ratios.
- **Maintenance Accountability:** The third-party provider is incentivized to keep the panels clean and the inverters operational to maximize their revenue, leading to high availability factors.

- **Data Granularity:** RESCO systems typically come with advanced cloud-based monitoring, which provided much of the high-resolution data for the 2023–2025 study period.

The synergy between these two models allows the campus to balance high-efficiency modern generation (RESCO) with institutional energy security (CAPEX).

2.2 Year-wise Generation Data

2.2.1 Analysis of Solar Generation - 2023

Table 2.1: Monthly Solar Generation (kWh) - 2023

Month	CAPEX	RESCO	Total
January	215,704	312,419	528,123
February	252,565	322,618	575,183
March	267,369	305,367	572,736
April	288,966	345,790	634,756
May	322,845	375,648	698,493
June	231,764	358,728	590,492
July	234,897	328,140	563,037
August	212,407	194,889	407,296
September	200,281	252,475	452,756
October	240,831	285,690	526,521
November	210,616	297,521	508,137
December	188,124	276,057	464,181
Total	2,834,369	3,687,342	6,521,711

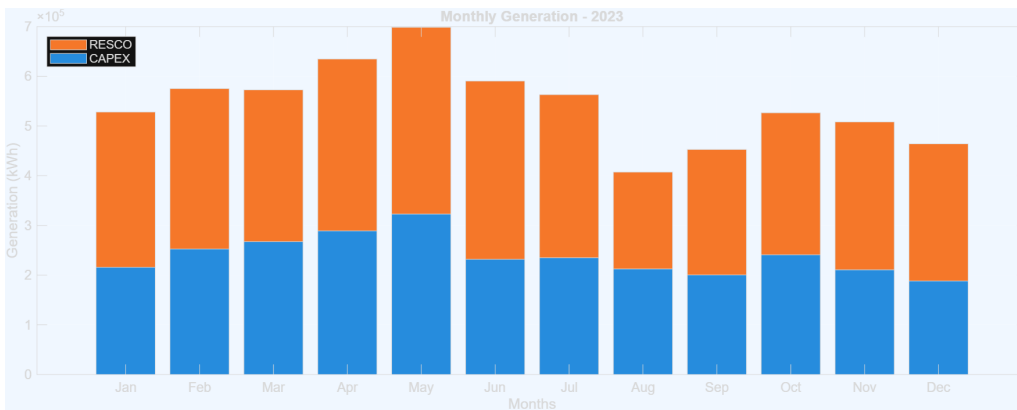


Figure 2.1: Visualizing the 2023 Generation Profile

2.2.2 Analysis of Solar Generation - 2024

Table 2.2: Monthly Solar Generation (kWh) - 2024

Month	CAPEX	RESCO	Total
January	218,804	189,117	407,921
February	202,813	326,059	528,872
March	253,772	345,640	599,412
April	257,517	357,110	614,627
May	247,997	346,813	594,810
June	227,256	319,608	546,864
July	203,371	266,690	470,061
August	194,179	257,671	451,850
September	197,669	277,308	474,977
October	192,449	254,501	446,950
November	176,410	264,878	441,288
December	177,291	244,032	421,323
Total	2,549,528	3,449,427	5,998,955

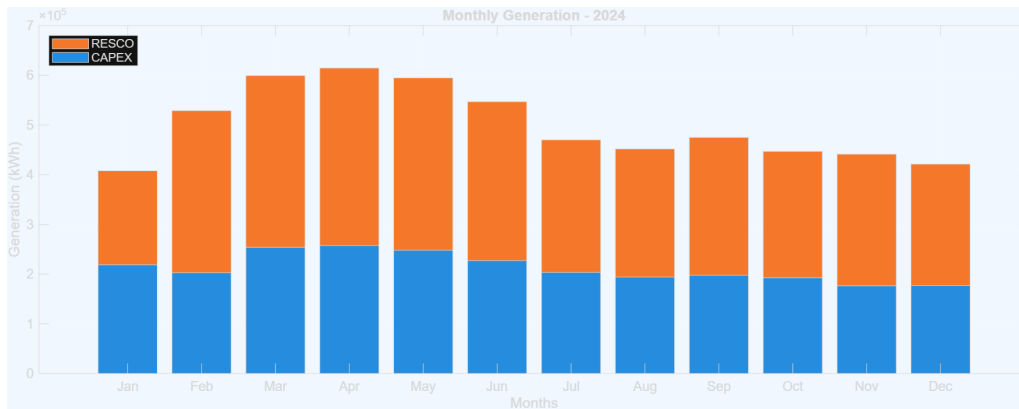


Figure 2.2: Visualizing the 2024 Generation Profile

2.2.3 Analysis of Solar Generation - 2025

Table 2.3: Monthly Solar Generation (kWh) - 2025

Month	CAPEX	RESCO	Total
January	192,595.0	261,506.0	454,101.0
February	202,732.0	280,416.0	483,148.0
March	239,865.0	383,180.0	623,045.0
April	258,543.3	343,668.7	602,212.0
May	260,501.4	362,533.3	623,034.7
June	191,076.9	279,556.6	470,633.5
July	183,121.0	256,825.8	439,946.8
August	198,223.0	269,899.7	468,122.7
September	210,441.9	301,433.3	511,875.2
October	204,142.0	299,141.0	503,283.0
November	211,899.4	304,147.1	516,046.5
December	165,500.0	239,492.7	404,992.7
Total	2,518,641	3,581,800	6,100,441

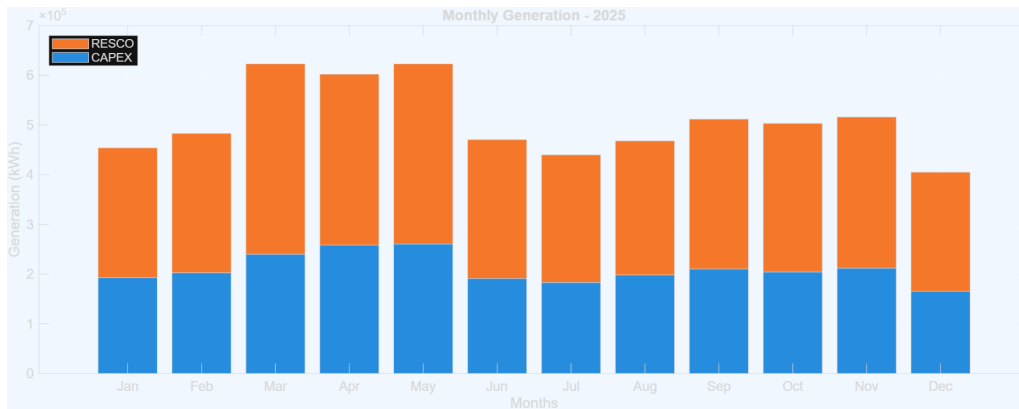


Figure 2.3: Visualizing the 2025 Generation Profile

2.3 Three-Year Comparison and Normalized Baseline

The study utilizes the aggregate of the previous three years to build a robust model for future deployment.

Table 2.4: Normalized Solar Generation Baseline (2023–2025)

Month	2023	2024	2025	Avg Monthly	Avg Daily
January	528,123	407,921	454,101	463,382	14,947.8
February	575,183	528,872	483,148	529,068	18,678.2
March	572,736	599,412	623,045	598,398	19,303.2
April	634,756	614,627	602,212	617,198	20,573.3
May	698,493	594,810	623,035	638,779	20,605.8
June	590,492	546,864	470,634	535,997	17,866.6
July	563,037	470,061	439,947	491,015	15,839.2
August	407,296	451,850	468,123	442,423	14,271.7
September	452,756	474,977	511,875	479,869	15,995.7
October	526,521	446,950	503,283	492,251	15,879.1
November	508,137	441,288	516,047	488,491	16,283.0
December	464,181	421,323	404,993	430,166	13,876.3
Annual Metric	6,521,711	5,998,955	6,100,441	517,253.1	17,010.0

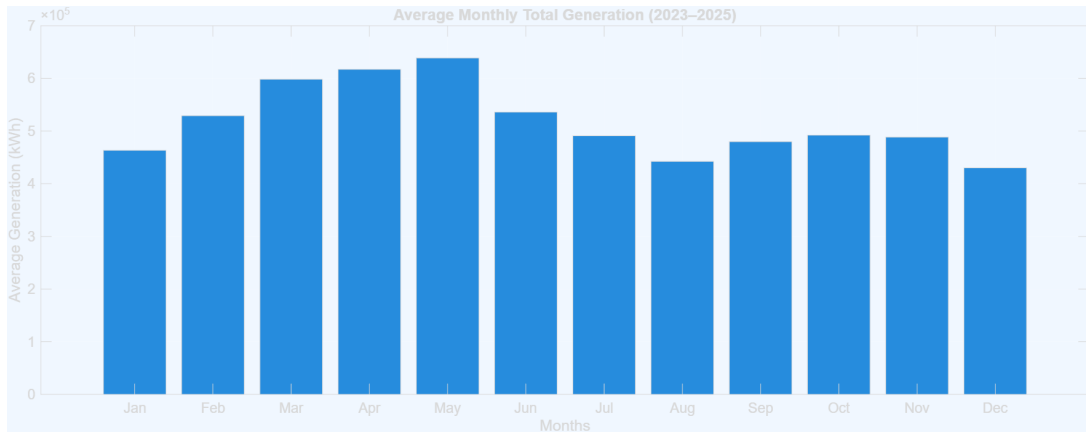


Figure 2.4: Aggregate Daily Generation Trends

2.4 Discussion and Trend Analysis: Solar Supply vs. Thermal Demand

Analyzing the averaged results in the context of the IIT Kharagpur academic calendar reveals a significant misalignment between peak solar supply and peak cooling demand. This analysis categorizes the year into three distinct operational phases for the framework.

2.4.1 The Surplus Phase (May – July)

The highest average generation occurs in May (20,605.8 kWh/day), with high yields continuing through July. However, this period coincides with the summer vacation, during which a large portion of the Halls of Residence are empty.

- **Implication:** The framework operates at maximum efficiency but minimum load.
- **Dashboard Utility:** During these months, the "Solar Utilization" KPI will remain extremely low, indicating a massive surplus that can be diverted back to the institutional grid or used for deep-cycle storage maintenance.

2.4.2 The Critical Design Window (August – October)

This is the most vital period for the framework. Students return to campus during the peak of the monsoon humidity, yet August suffers from a "Monsoon Dip" in solar yield (14,271.7 kWh/day) due to cloud cover.

- **Implication:** This represents the "Worst-Case Scenario" for the system.
- **Dashboard Utility:** The framework is designed around these constraints. To maintain comfort without exceeding the "Safe Harvest" limit, managers must use the CoP and TR/Room sliders to find an equilibrium that keeps solar utilization within sustainable bounds.

2.4.3 The Dormant Period (November – February)

From October to February, the ambient temperature in Kharagpur drops significantly, rendering residential cooling unnecessary. While solar yield also dips during the winter solstice in December (13,876.3 kWh/day), the lack of thermal demand means the system remains dormant.

- **Implication:** The 5.5 MWp solar infrastructure acts as a standard grid-tied system during this period, supporting institutional laboratory and administrative loads rather than residential cooling.

2.4.4 The Transition Phase (March – April)

As temperatures rise in the spring, cooling demand resumes. With high solar elevation angles and clear skies, March and April provide an ideal balance of high yield (approx. 19,300–20,500 kWh/day) and high occupancy. This phase allows for the full utilization of the **Waterfall Tiered** 2-3-4 hour cooling schedule with minimal grid dependence.

Chapter 3

Baseline Demand Modeling

3.1 Methodology for Dynamic Energy Estimation

The second phase of this framework focuses on establishing a Baseline Demand Model using the three currently AC-equipped residences: SBP-1, SBP-2, and ABVP. Unlike static models, this phase treats environmental and mechanical factors as variables. By parameterizing these values, the framework can be calibrated for different hardware efficiencies before scaling the logic to the 21-hall campus proposal.

3.2 Mathematical Framework for Power and Energy

To account for the significant variance in capacity across the 21 Halls of Residence, the transition from thermal cooling requirements to electrical consumption is governed by variables calculated on a per-room basis:

- **Total Aggregate Rooms (N_{rooms}):** The sum of rooms in the selected halls (e.g., 900 rooms for the baseline study).
- **Cooling Intensity (TR_{room}):** The thermal load required per room (Adjustable: 0.1 to 1.0 TR).
- **Efficiency (CoP):** The Coefficient of Performance of the chiller plants (Adjustable: 2.0 to 5.0).
- **Duration (t):** The length of the cooling window, determined by the **Waterfall Tiered** logic or **Manual Allocation**.

3.2.1 Step 1: Total Cooling Capacity (TR_{total})

The aggregate thermal load for the selected halls is the product of the total room count and the cooling intensity:

$$TR_{total} = N_{rooms} \times TR_{room} \quad (3.1)$$

3.2.2 Step 2: Instantaneous Electrical Power Demand (P_{elec})

Using the thermal equivalent ($1 \text{ TR} \approx 3.517 \text{ kW}_{thermal}$), the instantaneous electrical power required is calculated as:

$$P_{elec}(kW) = \frac{TR_{total} \times 3.517}{CoP} \quad (3.2)$$

3.3 Validation Case: Baseline Waterfall Calculations

To validate the mathematical framework and maintain consistency with the campus baseline, we apply a specific configuration used for the **Waterfall Tiered** logic. This case study utilizes the baseline halls (SBP-1, SBP-2, and ABVP) which have a combined capacity of 900 rooms ($N_{rooms} = 900$). We assume a benchmark intensity of 0.6 TR/room and a system efficiency (CoP) of 3.2.

3.3.1 Instantaneous Load Calculation

Substituting the baseline values into the framework, the instantaneous electrical demand is:

$$P_{elec} = \frac{(900 \times 0.6) \times 3.517}{3.2} \approx 594 \text{ kW} \quad (3.3)$$

3.3.2 Energy Consumption for 24/7 vs. 4-Hour Operation

To illustrate the efficiency of the scheduled window, we calculate the energy consumption for both continuous and limited operations for these 900 rooms:

- **Continuous (24/7) Operation:**

$$E_{24/7} = 594 \text{ kW} \times 24 \text{ hours} = 14,256 \text{ kWh} \quad (3.4)$$

- **Scheduled (4-Hour) Operation:**

$$E_{4hr} = 594 \text{ kW} \times 4 \text{ hours} = 2,376 \text{ kWh} \quad (3.5)$$

3.4 Comparative Load Analysis

The table below contrasts these requirements, highlighting the significant reduction in daily energy demand achieved through intelligent scheduling.

Table 3.1: Comparative Operational Metrics: 24/7 vs. Scheduled Window

Metric	4-Hour Window	24/7 Operation
Electrical Power Demand	594 kW	594 kW
Cumulative Daily Energy	2,376 kWh	14,256 kWh
Avg. Solar Coverage Required	13.97%	83.81%
Min. Solar Coverage Required (Aug)	17.12%	102.73%

3.5 Solar Coverage Implications

By shifting the load to a 4-hour window, the "Solar Coverage Required" drops from a staggering 83.81% to just 13.97%. This reduction is the fundamental driver that makes the framework viable across the entire campus. It is important to note that thermal demand is seasonal; from November to February, ambient temperatures drop and cooling is not required, allowing the solar infrastructure to support institutional loads exclusively.

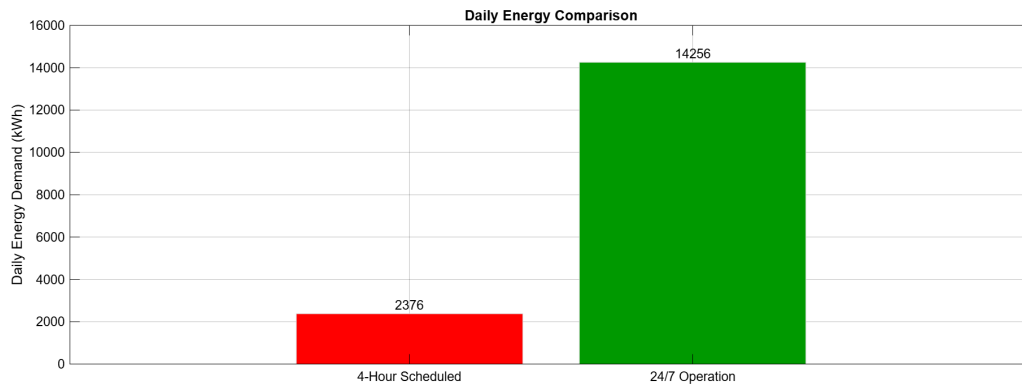


Figure 3.1: Energy Demand Comparison: Scheduled Window vs. Continuous Operation

3.6 Visualization of Hourly Load Profile

The hourly load profile demonstrates the "Square Wave" nature of our scheduled demand. Unlike institutional loads that follow the sun's curve, our demand is a discrete block of energy shifted to the midnight hours.

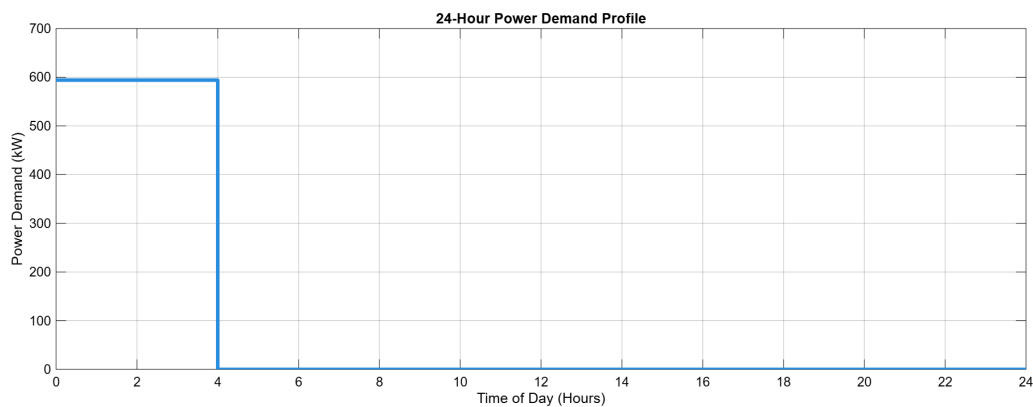


Figure 3.2: Baseline Hourly Power Demand (12 AM–4 AM)

Chapter 4

Storage Requirement Mapping and Scenario Analysis

4.1 Mapping Available Renewable Energy

The primary objective of this phase is to quantify the "Daily Solar Surplus"—the portion of the 5.5 MWp generation that can be diverted from daytime institutional loads to be stored for night-time cooling. Given that the campus generates approximately 6 GWh annually, the framework identifies the seasonal "Safe Harvest" limit.

This limit represents the maximum energy that can be extracted during peak sun hours (10 AM – 3 PM) without compromising the essential daytime operations of the departments and laboratories. Based on the 2023–2025 datasets, the framework is designed to optimize this surplus, specifically accounting for the constraints of the monsoon humidity in August when demand is high but solar yield dips.

4.2 Variable Storage Scaling and Load Coverage ($X\%$)

To avoid the prohibitive costs of oversized hardware, the framework utilizes a dynamic "Load Coverage Factor" ($X\%$), controlled via the dashboard slider. The Design Storage Capacity (S_{design}) is calculated based on the total aggregate rooms selected by the manager.

4.2.1 The Scaling Formula

The required storage capacity is governed by the following mathematical relationship:

$$S_{design}(kWh) = X\% \times (N_{rooms} \times P_{elec_room} \times t) \quad (4.1)$$

Where:

- $X\%$: The Load Coverage Factor (User-adjustable from 10% to 100%).
- N_{rooms} : Total aggregate rooms across all active halls.
- P_{elec_room} : Electrical demand per room (derived from user-selected CoP and TR).
- t : Duration of the cooling window (**Waterfall Tiered** or **Manual Allocation**).

4.2.2 Example Case: Waterfall Baseline Storage Requirements

To validate the scaling requirements, we utilize the **Waterfall Tiered** baseline for 900 rooms across SBP-1 and SBP-2. In this scenario ($P_{elec.room} \approx 0.66$ kW), we evaluate the storage needed across varying durations and coverage targets.

Table 4.1: Storage Capacity Matrix for 900 Rooms ($P_{total} = 594$ kW)

Duration (t)	Total Consumption (E_{total})	Storage @ 75%	Storage @ 50%
2 Hours	1,188 kWh	891 kWh	594 kWh
3 Hours	1,782 kWh	1,336.5 kWh	891 kWh
4 Hours	2,376 kWh	1,782 kWh	1,188 kWh

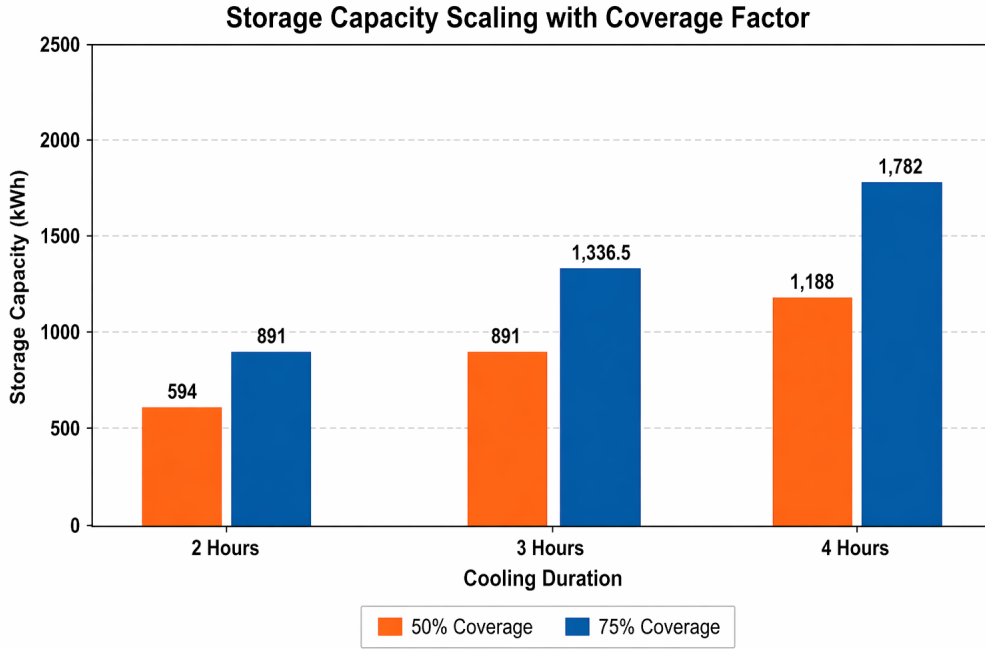


Figure 4.1: Storage Capacity Requirements across Different Coverage Factors

4.3 Scaling Relationships: Room Support vs. Coverage

The framework establishes a dynamic relationship between stored energy and service breadth. If the available solar surplus is low, the software evaluates trade-offs in real-time. The maximum number of rooms (N_{rooms}) supported by a fixed storage volume ($S_{available}$) is:

$$N_{rooms} = \frac{S_{available}}{X\% \times P_{elec.room} \times t} \quad (4.2)$$

This formula is the engine behind the scheduling algorithm, ensuring that even under **Manual Allocation** settings, the energy discharge never exceeds hardware safety limits.

4.4 Evaluation of Storage Strategies: BESS vs. TES

The framework evaluates two primary technological pathways for time-shifting solar energy:

4.4.1 Battery Energy Storage Systems (BESS)

BESS utilizes chemical storage (Lithium-Ion) to store DC power. While offering high efficiency (85-90%), BESS is constrained by high capital expenditure and a limited cycle life at deep discharge rates.

4.4.2 The Proposed Solution: Thermal Energy Storage (TES)

This project proposes **Thermal Energy Storage (TES)** as the optimal pathway for the IIT Kharagpur campus. TES involves running chillers during peak solar hours to produce chilled water, which is stored in insulated tanks and circulated during the midnight window.

- **Economic Viability:** Significantly lower cost per kWh for large-scale thermal shifts.
- **Integration:** Highly compatible with the existing centralized chiller plumbing in halls like SBP and ABVP.

4.5 TES Sizing: Chilled Water Storage Calculations

To demonstrate the feasibility of the TES proposal, we calculate the required water volume (V) to store the campus-wide energy target ($S_{design} \approx 9,138$ kWh).

$$V(m^3) = \frac{S_{design} \times 3600}{\rho \times c_p \times \Delta T} \quad (4.3)$$

Where:

- ρ : Density of water ($1000 \text{ kg}/m^3$).
- c_p : Specific heat of water ($4.186 \text{ kJ}/kg \cdot K$).
- ΔT : Temperature difference (assumed as 5 K for standard chiller loops).

Substituting the campus-wide values:

$$V = \frac{9,138 \times 3600}{1000 \times 4.186 \times 5} \approx 1,572 \text{ m}^3 \quad (4.4)$$

This volume is equivalent to a storage tank approximately 20m in diameter and 5m in height, confirming that TES is a technologically viable and physically scalable solution for the proposed 21-hall rollout.

4.6 Impact on Final Framework Design

The conclusion derived from this storage analysis underscores the importance of the partial Load Coverage Factor ($X\%$) as the cornerstone of a realistic engineering solution for the IIT Kharagpur campus. By intentionally sizing the storage system below the absolute theoretical maximum required for 100% autonomous operation, we effectively circumvent the “Over-Engineering Trap”.

Designing a system for 100% autonomy would necessitate a storage footprint that is both financially prohibitive and physically impossible to accommodate within the existing campus landscape. Instead, the framework’s reliance on a variable $X\%$ factor (optimized at 0.60 for the campus baseline) provides a more sustainable and adaptable design for the following reasons:

- **Economic Viability:** Reducing the storage capacity requirement by 40% significantly lowers the initial Capital Expenditure (CAPEX). This makes the 21-hall rollout more attractive for institutional funding by focusing on the 80% of the time where solar yields are optimal, rather than over-spending for the 20% of extreme weather outliers.
- **Physical Footprint and Safety:** Utilizing Thermal Energy Storage (TES) tanks at a partial scaling factor ensures that the storage infrastructure remains within a manageable size (approx. $1,572\text{ m}^3$). This prevents the need for multiple massive installations that would disrupt campus greenery and pedestrian pathways.
- **Grid-Solar Synergy:** The design acknowledges that the campus grid is not an adversary but a stabilizer. By using the grid to handle peak transients and the remaining 40% of the load, the system ensures that the centralized chillers do not suffer from voltage fluctuations that might otherwise occur in a purely off-grid solar-storage setup.
- **Operational Flexibility via UI:** By embedding this $X\%$ logic directly into the Streamlit dashboard sliders, the manager can adjust the framework’s behavior in real-time. This allows for “Dynamic Budgeting”—where the system can be pushed to higher coverage during peak summer months and throttled back during the August monsoon dip to maintain sustainability.

Ultimately, this design choice transforms the project from an idealized academic model into a robust Decision Support System. It demonstrates that with intelligent scheduling and a calculated hybrid energy blend, meaningful night-time cooling can be provided to the entire student community using a technologically scalable and financially responsible hardware footprint.

Chapter 5

Scheduling Algorithm and Campus Scaling

5.1 The Waterfall Tiered Preset: A Baseline for Feasibility

While the software framework provides a **Manual Allocation** for granular control, it is essential to establish a "Standardized Preset" to evaluate the feasibility of providing cooling to the entire 21-hall campus proposal. The "**Waterfall Tiered Model**" serves as this baseline. By categorizing the 8,173 rooms at IIT Kharagpur into a multi-tiered system based on capacity, we can demonstrate that equitable cooling is possible within the limits of the existing 5.5 MWp solar infrastructure.

5.2 Multi-Tiered Campus Categorization

The categorization is based on the total room count of each hall. This ensures that the instantaneous load on the storage system is managed effectively through an "Inverse Duration" principle.

5.2.1 Inverse Allocation Logic

In this baseline preset, smaller halls (low aggregate room count) are allocated a longer duration of 4 hours. Conversely, the largest "load giants" are allocated a concentrated 2-hour window. This prevents the rapid depletion of the Thermal Energy Storage (TES) while ensuring every student receives meaningful nocturnal relief.

5.3 Theoretical Energy Consumption and Storage Scaling

The total energy required (E_{total}) for this baseline cycle is calculated by aggregating the power requirements of all 8,173 rooms over their respective tiered durations.

Table 5.1: Tiered Categorization and Room-Based Power Demand

Category	Room Range	Duration	Halls included	Power (P_{tier})
Tier 1: Small	< 270 rooms	4 Hours	Gokhale, SNVH, BC Roy, HJB, MT, SNIG, RLB, JCB	914.1 kW
Tier 2: Medium	270–600 rooms	3 Hours	Patel, Nehru, LLR, VS, MS, RK, Azad, SBP-1, SBP-2, RP	2,612.9 kW
Tier 3: Large	> 600 rooms	2 Hours	LBS, MMM, BRH	1,867.1 kW
CAMPUS TOTAL	8,173 Rooms	Variable	21 Halls of Residence	5,394.1 kW

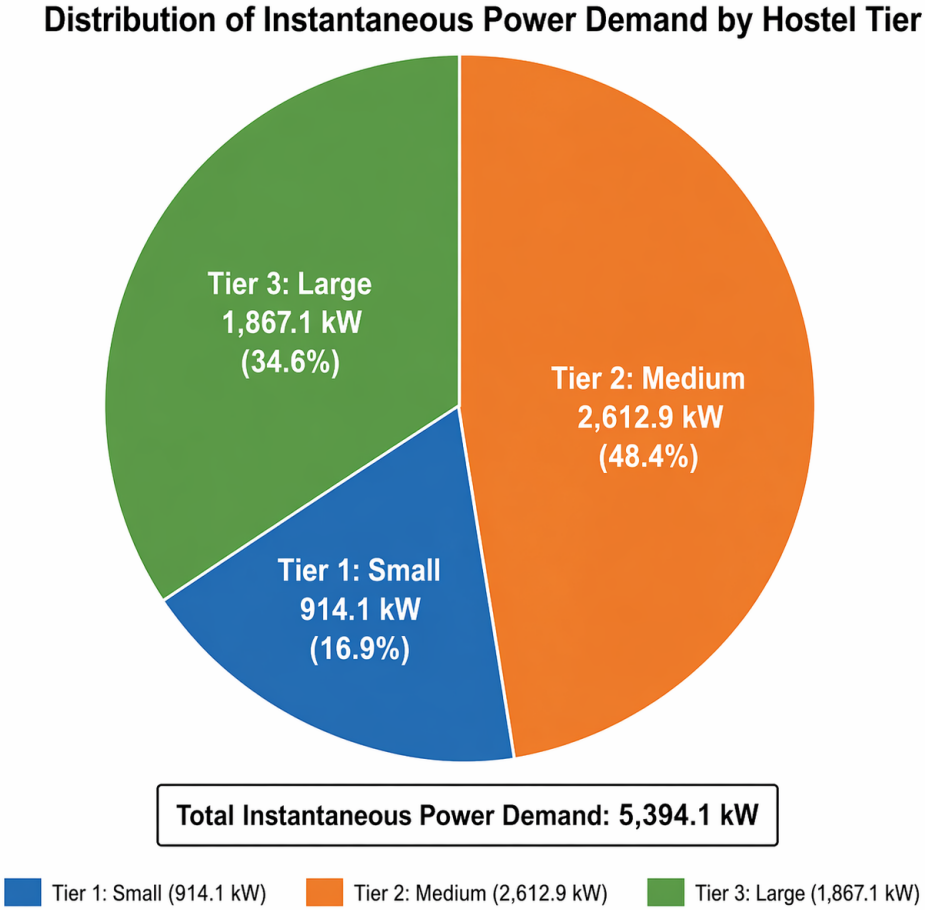


Figure 5.1: Distribution of Instantaneous Power Demand by Tier

5.3.1 Total Energy Requirement (E_{total})

The cumulative energy consumption is derived as:

$$E_{total} = (P_{T1} \times 4) + (P_{T2} \times 3) + (P_{T3} \times 2) \quad (5.1)$$

Using the per-room power constants, this totals:

$$E_{total} = (914.1 \times 4) + (2612.9 \times 3) + (1867.1 \times 2) = 15,229.5 \text{ kWh} \quad (5.2)$$

5.3.2 Optimal Storage Scaling (The 0.60 X-Factor)

As established in the dashboard’s default settings, the framework adopts a 0.60 scaling factor. The storage system is sized to handle 60% of the room-based load, with the remaining 40% supplemented by the grid to stabilize chiller transients.

$$S_{design} = 0.60 \times 15,229.5 \text{ kWh} = 9,137.7 \text{ kWh} \quad (5.3)$$

5.3.3 Sustainability Metric: Solar Utilization

Comparing this baseline energy requirement to the average daily solar generation (17,010 kWh), the utilization rate is:

$$\text{Utilization} = \frac{9,137.7 \text{ kWh}}{17,010 \text{ kWh}} \approx 53.7\% \quad (5.4)$$

This confirms that even with all 8,173 rooms active, the proposal consumes only about half of the solar surplus, leaving ample energy for daytime institutional needs.

5.4 The Waterfall Scheduling Strategy

The algorithm manages the baseline load through a ”Sequential Waterfall” strategy. This prevents a massive ”cliff-drop” in power demand at the end of the cooling window.

5.4.1 Sequential Tiering Phases

The baseline cooling window is executed in three distinct phases:

- **Phase 1 (12 AM – 2 AM):** Full campus coverage. All 8,173 rooms are active. Total load is 5,394.1 kW.
- **Phase 2 (2 AM – 3 AM):** Tier 3 (Large Halls) is disconnected. Tier 1 and Tier 2 rooms continue.
- **Phase 3 (3 AM – 4 AM):** Tier 2 (Medium Halls) is disconnected. Tier 1 rooms complete their final hour.

5.5 Scaling Analysis and Trade-offs

By analyzing this baseline, the framework highlights the critical trade-off between coverage and duration. A ”Flat Schedule” where all 8,173 rooms receive 4 hours of cooling would require over 22,000 kWh—a figure that exceeds the daily solar supply. The **Waterfall Tiered** preset proves that by utilizing variable durations, the proposal remains both economically viable and technologically scalable.

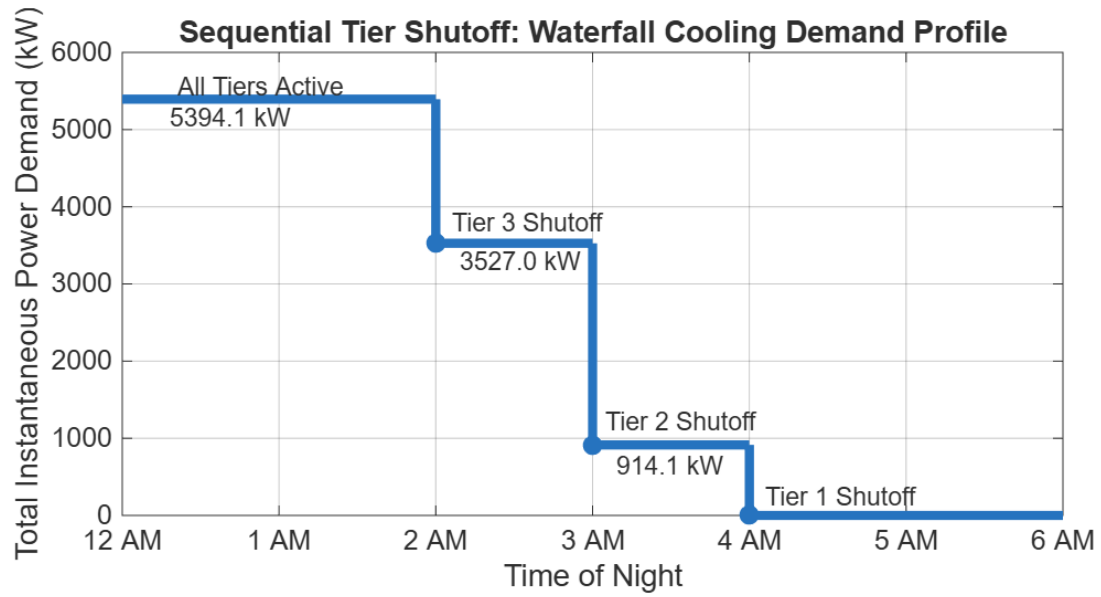


Figure 5.2: Tiered Sequential Cooling Schedule (12 AM – 4 AM)

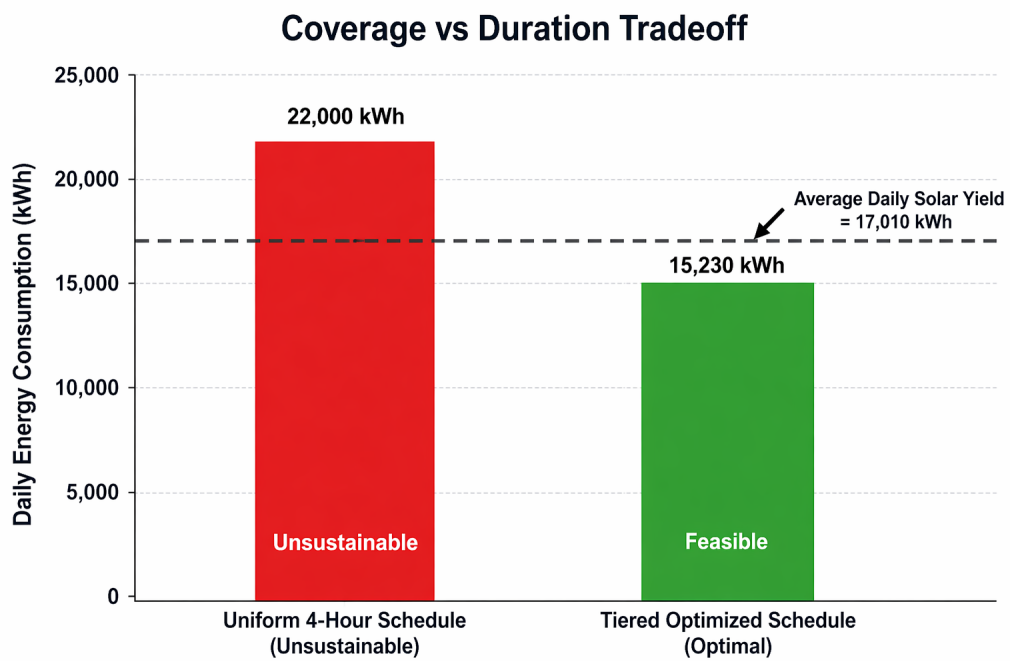


Figure 5.3: Trade-off Analysis: Tiered Scheduling vs. Uniform Scheduling

Chapter 6

User Interface Development (Python/Streamlit)

6.1 Objective: Transitioning from Data to Decision Support

The final phase of the project involves the translation of the solar resource assessment, room-based demand modeling, and tiered scheduling logic into a functional, interactive software tool. While the mathematical framework establishes feasibility, an institutional administrator requires a granular interface to manage energy dispatch in real-time.

Utilizing the Python-based Streamlit library, we developed a "Solar-to-Cooling Dashboard". This updated interface serves as a dynamic Decision Support System (DSS). It allows campus energy managers to tune physical parameters—such as chiller efficiency and cooling intensity—to see the real-time impact on the campus energy budget.

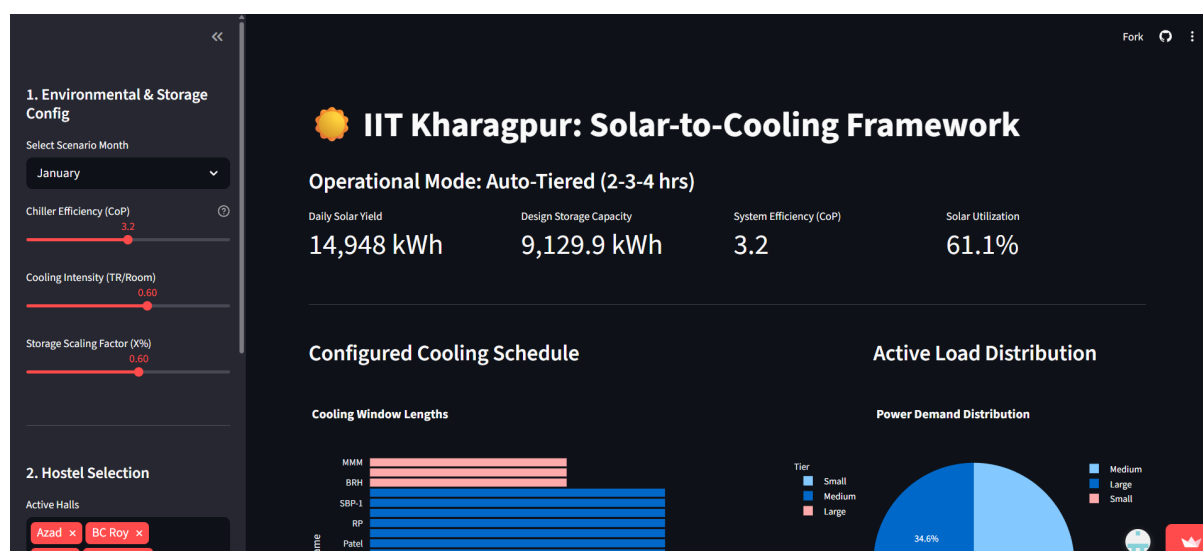


Figure 6.1: Primary Interface of the IIT Kharagpur Solar-to-Cooling Framework

6.2 Dashboard Architecture and Control Modules

The framework is structured into three primary control modules in the sidebar, designed to provide the manager with full operational flexibility.

6.2.1 Environmental and Storage Configuration

This module allows the user to define the physical and seasonal constraints of the system:

- **Scenario Month:** Fetches the normalized 2023–2025 daily yield data to ground the simulation in reality.
- **Chiller Efficiency (CoP):** A variable slider (2.0 to 5.0) that replaces static efficiency values, allowing for hardware-specific calibration.
- **Cooling Intensity (TR/Room):** A slider (0.1 to 1.0 TR) to adjust the thermal comfort level per room based on current occupancy or ambient humidity.
- **Storage Scaling Factor ($X\%$):** Adjusts the percentage of the cooling load to be met by stored solar energy versus the grid supplement.

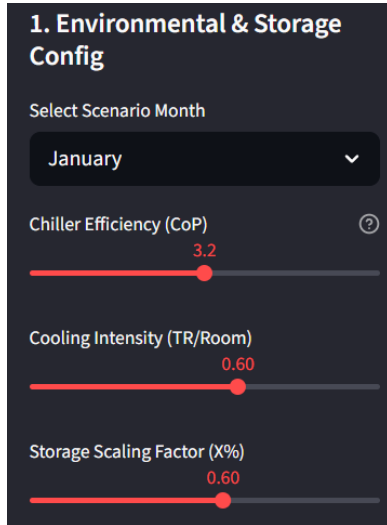


Figure 6.2: Environmental and Storage Config

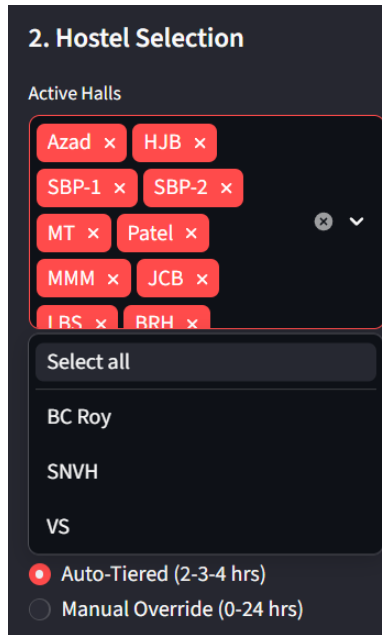


Figure 6.3: Hostel Selection / Selective Dispatch

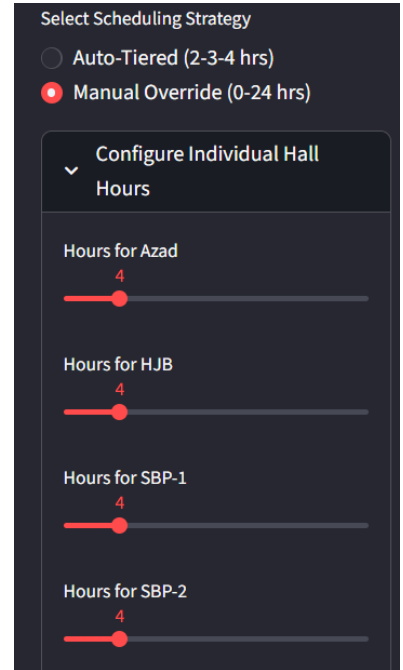


Figure 6.4: **Manual Allocation** Duration Sliders

6.2.2 Hostel Selection (Selective Dispatch)

The interface features a multi-select tool for the 21 Halls of Residence. This allows for "Selective Dispatch," where an administrator can manually toggle specific halls to provide cooling based on room availability or maintenance schedules. The system automatically aggregates the total room count (N_{rooms}) of the selected halls to recalculate power demand.

6.2.3 Scheduling Strategy: Waterfall Tiered vs. Manual Allocation

The manager can choose between preset and manual modes:

1. **Waterfall Tiered (2-3-4 hrs):** Implements the waterfall preset for campus-wide equity.
2. **Manual Allocation (0-24 hrs):** Provides individual sliders for every selected hall, allowing the manager to define custom cooling durations based on specific residential needs or exam schedules.

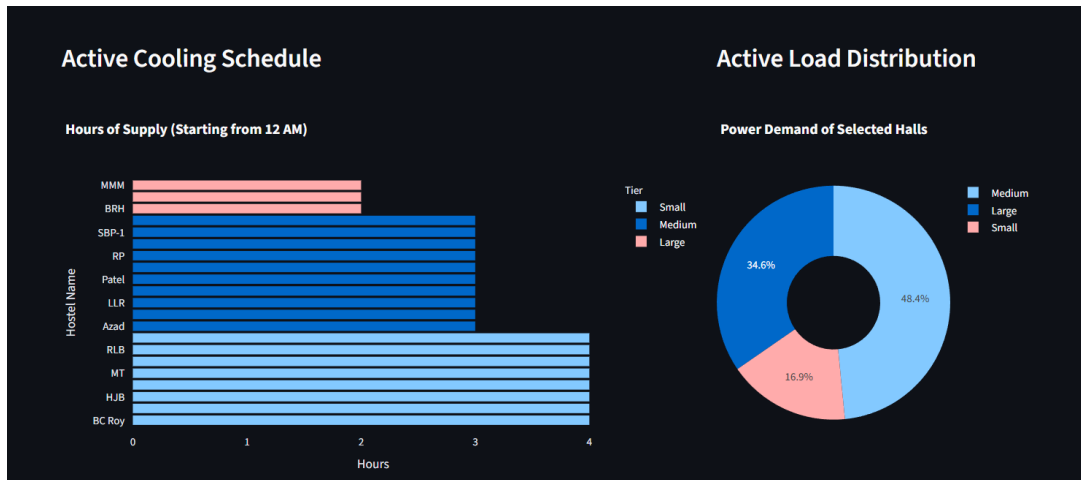


Figure 6.5: Active Cooling Schedule and Load Distribution Visualization

6.3 Key Visualizations and Decision Support

The dashboard utilizes advanced data visualization to simplify the complexity of energy blending and room-based load distribution.

6.3.1 Power Supply Analysis (Stacked Bar Chart)

The "Instantaneous Power Source" chart visualizes the blending of energy sources. It shows the "Solar-Storage Contribution" (the X factor) stacked against the "Grid Supplement". This proves that the framework is a realistic engineering solution that uses the grid as a stabilizer for chiller transients rather than an idealized off-grid scenario.

6.3.2 Sustainability Metrics (KPIs)

At the top of the dashboard, four "Key Performance Indicators" (KPIs) provide instant feedback on the feasibility of the current configuration:

1. **Daily Solar Yield:** Displays the real-time average daily generation for the selected scenario month based on the 2023–2025 normalized dataset.

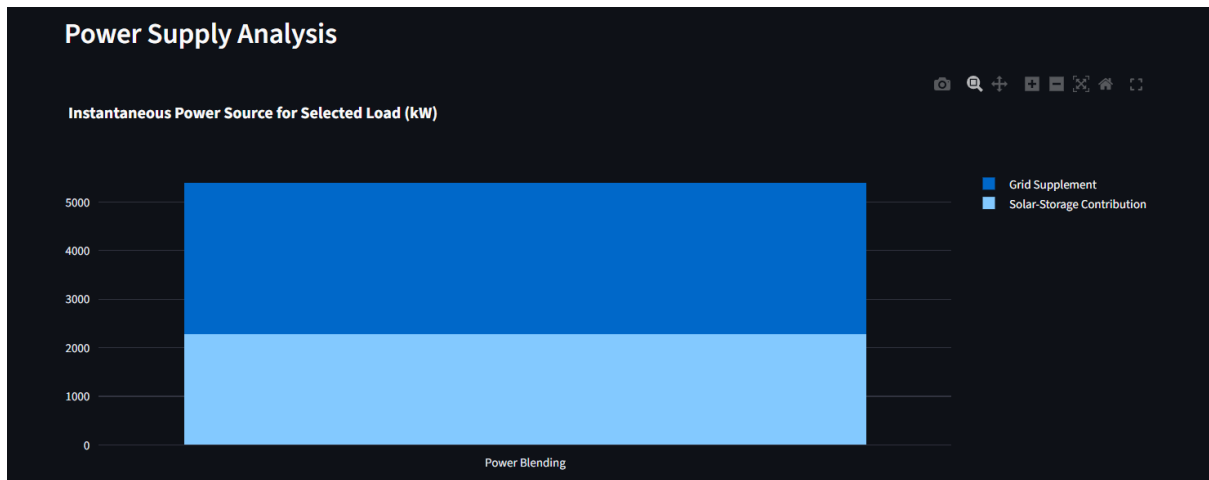


Figure 6.6: Stacked Power Supply Analysis: Solar-Storage vs. Grid Supplement

2. **Design Storage Capacity:** Dynamically recalculates the required kWh of energy storage based on the total room count, selected cooling intensity, and duration.
3. **System Efficiency (CoP):** Reflects the currently active hardware efficiency benchmark. This metric is critical as it defines the ratio of thermal cooling provided per unit of electrical energy consumed, directly influencing the power demand calculations.
4. **Solar Utilization:** The most vital safety check of the framework. It calculates the percentage of the daily solar yield consumed by the selected cooling schedule. If this value exceeds a sustainable threshold (e.g., 90%), the system alerts the manager that they are over-budgeting the available solar energy.

Operational Mode: Auto-Tiered (2-3-4 hrs)			
Daily Solar Yield	Design Storage Capacity	System Efficiency (CoP)	Solar Utilization
14,948 kWh	9,129.9 kWh	3.2	61.1%

Figure 6.7: Dashboard Header displaying primary Key Performance Indicators (KPIs)

6.4 Framework Accessibility and Live Deployment

To facilitate real-time testing and institutional evaluation, the complete framework has been deployed as a standalone web application. The live dashboard can be accessed at:

<https://iitkgp-cooling-dashboard.streamlit.app/>

This ensures that the data-driven scheduling logic is accessible to campus administrators without the need for local environment configuration.

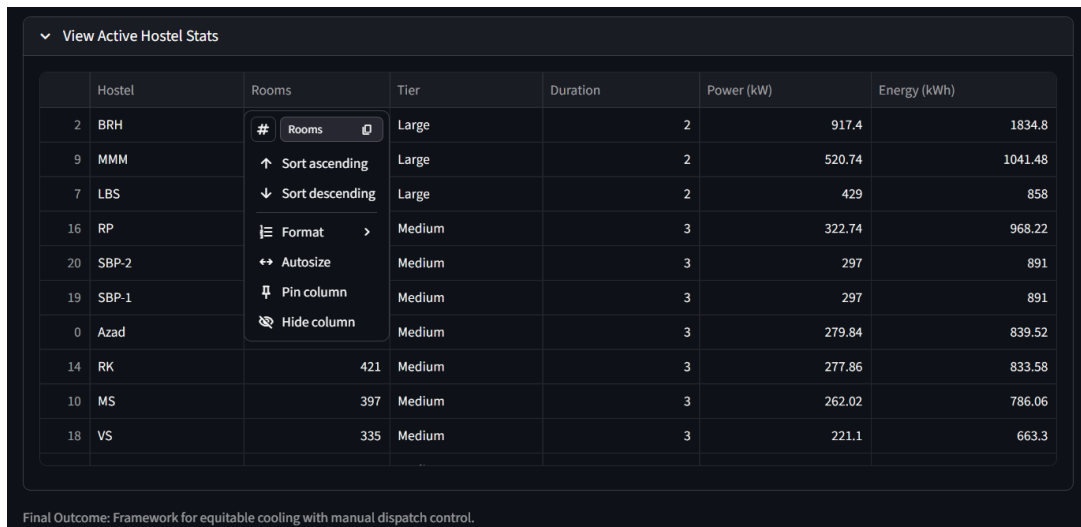


Figure 6.8: Granular Active Hall Statistics and Energy Consumption Table

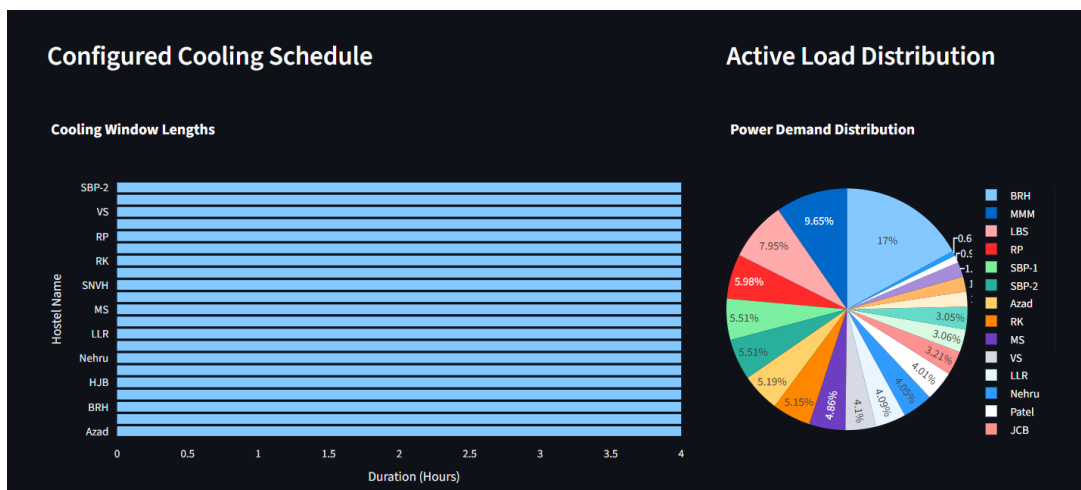


Figure 6.9: Cooling Schedule and Load Distribution for Manual Allocation

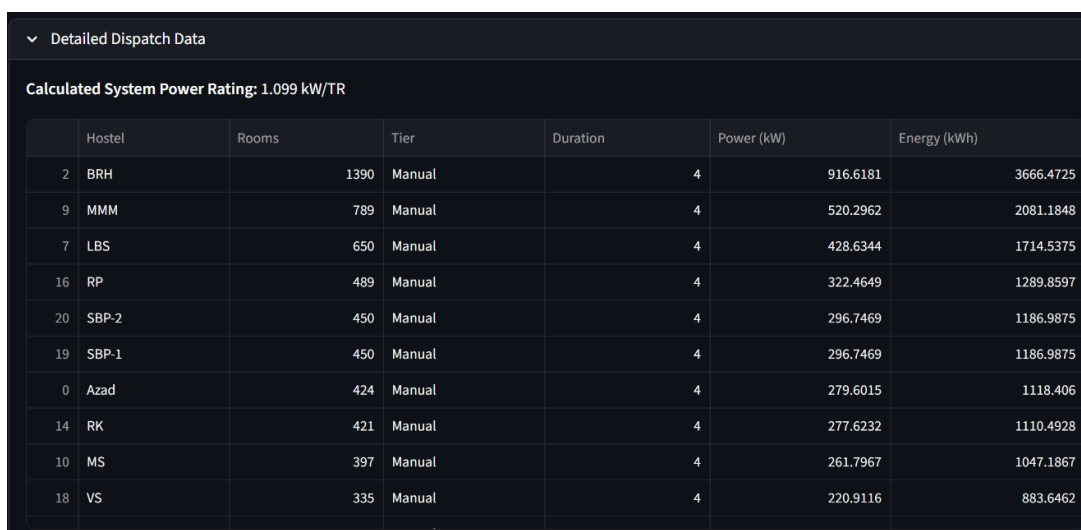


Figure 6.10: Active Hall Statistics and Consumption Table for Manual Allocation

Chapter 7

Conclusion

This project establishes a comprehensive, software-driven framework for achieving institutional sustainability at IIT Kharagpur through solar-powered night-time cooling. By creating a data-driven bridge between the campus’s 5.5 MWp solar PV infrastructure and the residential demand of 21 Halls of Residence, the framework proves that significant energy time-shifting is both technologically feasible and economically viable.

The successful development of this system provides several key insights into campus energy management:

- **Dynamic Resource Optimization:** Through a granular analysis of solar generation data from 2023–2025, the project identifies the seasonal "Safe Harvest" limits. This ensures that cooling is provided without cannibalizing the energy required for daytime academic and laboratory operations.
- **Variable Thermodynamic Modeling:** By opting for a room-based demand model, the framework allows for real-time calibration of physical parameters such as Chiller Efficiency (CoP) and Cooling Intensity (TR/Room). This flexibility ensures the system remains accurate across varying hardware conditions and environmental demands.
- **Scalable Infrastructure:** The evaluation of storage strategies identifies Thermal Energy Storage (TES) as the optimal pathway for a campus-wide rollout. Sizing calculations confirm that a 1,572 m^3 chilled water storage system is a physically realistic and cost-effective solution for supporting all 8,173 rooms.
- **Operational Equity and Control:** The implementation of the **Waterfall Tiered** algorithm ensures an equitable distribution of energy, while the **Manual Allocation** functionality provides energy managers with the granular control necessary for selective dispatch and priority scheduling.

The results indicate that providing 2–4 hours of nocturnal cooling to the entire campus proposal is sustainable, requiring approximately 53.7% of the daily solar surplus. The culmination of this work into a functional Streamlit dashboard transitions the project from a theoretical study to a live Decision Support System (DSS). Ultimately, this framework provides a transparent and scalable blueprint for IIT Kharagpur to maximize its renewable energy potential while significantly improving the thermal comfort and quality of life for its residential community.